

Solid state Chapter 3 homework

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Problem 1

Question 1

The generalized eigenvalue problem

$$H\mathbf{c} = E S\mathbf{c}$$

was solved numerically in Python using the given Hamiltonian and overlap matrices.

The resulting energies are

$$E_1 = -6.05084746 \text{ eV}, \quad E_2 = -4.31707317 \text{ eV}.$$

The corresponding eigenvectors are

$$\mathbf{c}_1 = \begin{pmatrix} -0.65094455 \\ -0.65094455 \end{pmatrix}$$

and

$$\mathbf{c}_2 = \begin{pmatrix} 0.78086881 \\ -0.78086881 \end{pmatrix}.$$

Question 2

The energies obtained in Question 1 were already returned sorted:

$$E_1 = -6.05084746 \text{ eV}, \quad E_2 = -4.31707317 \text{ eV}.$$

Because the basis is nonorthogonal, the eigenvectors must satisfy the normalization condition

$$\mathbf{c}^T S \mathbf{c} = 1.$$

For the first eigenvector,

$$\mathbf{c}_1^T S \mathbf{c}_1 = 1.0000000000000002 \approx 1,$$

and for the second eigenvector,

$$\mathbf{c}_2^T S \mathbf{c}_2 = 1.0.$$

Therefore, both eigenvectors were already properly normalized with respect to the overlap matrix S .

Question 3

Using

$$E_+ = \frac{\epsilon_s + h}{1 + s}, \quad E_- = \frac{\epsilon_s - h}{1 - s}.$$

and

$$\epsilon_s = -5.34 \text{ eV}, \quad h = -1.80 \text{ eV}, \quad s = 0.18,$$

gives

$$E_+ = \frac{-5.34 - 1.80}{1 + 0.18} = -6.05084746 \text{ eV},$$

and

$$E_- = \frac{-5.34 + 1.80}{1 - 0.18} = -4.31707317 \text{ eV}.$$

These values agree with the numerical results.

The lower-energy eigenvector is

$$\mathbf{c}_1 = \begin{pmatrix} -0.65094455 \\ -0.65094455 \end{pmatrix}.$$

It has equal coefficients with the same sign, so it is the symmetric bonding state.

The higher-energy eigenvector is

$$\mathbf{c}_2 = \begin{pmatrix} 0.78086881 \\ -0.78086881 \end{pmatrix}.$$

It has equal coefficients with opposite signs, so it is the antisymmetric antibonding state.

Question 4

Using the orthogonal approximation $S = I$ while keeping the Hamiltonian unchanged, the eigenvalue problem becomes

$$H\mathbf{c} = E\mathbf{c}.$$

The orthogonal-approximation energies obtained numerically are

$$E_1^{(S=I)} = -7.14 \text{ eV}, \quad E_2^{(S=I)} = -3.54 \text{ eV}.$$

With overlap $s = 0.18$, the energies were

$$E_1^{(\text{overlap})} = -6.05084746 \text{ eV}, \quad E_2^{(\text{overlap})} = -4.31707317 \text{ eV}.$$

The energy shifts caused by overlap were calculated as

$$\Delta E = E_{\text{overlap}} - E_{S=I}.$$

Therefore,

$$\Delta E_1 = -6.05084746 - (-7.14) = +1.08915254 \text{ eV},$$

and

$$\Delta E_2 = -4.31707317 - (-3.54) = -0.77707317 \text{ eV}.$$

Thus, including the finite overlap shifts the lower-energy state upward by 1.08915254 eV and the higher-energy state downward by 0.77707317 eV.

Question 5

The overlap parameter was varied from

$$0.00 \leq s \leq 0.40$$

while keeping ϵ_s and h fixed.

The two energy branches were evaluated as

$$E_+(s) = \frac{\epsilon_s + h}{1 + s}, \quad E_-(s) = \frac{\epsilon_s - h}{1 - s}.$$

Figure 1 shows the resulting energies as functions of s . The assigned value $s = 0.18$ is marked by the vertical dashed line.

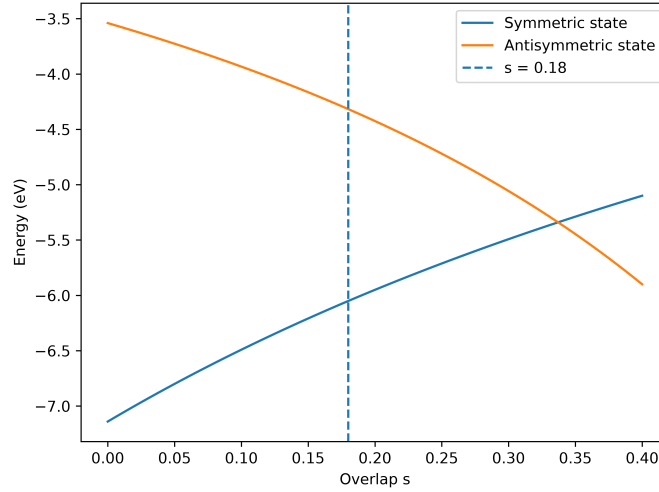


Figure 1: Energy levels as a function of the overlap parameter s , with ϵ_s and h held fixed. The dashed line marks $s = 0.18$.

As s increases over the assigned range, the symmetric-state energy moves upward while the antisymmetric-state energy moves downward, so the two energy levels approach each other.

Question 6

Because the localized orbitals are nonorthogonal, their overlap

$$S_{ij} = \langle i | j \rangle$$

is not equal to the identity matrix. When the Schrödinger equation is projected onto this basis, the overlap terms remain and gives the generalized eigenvalue problem

$$H\mathbf{c} = ES\mathbf{c}.$$

Replacing S by the identity matrix I assumes that the basis orbitals are orthogonal, so that the off-diagonal overlaps vanish. Since the actual overlap in this problem is $s = 0.18 \neq 0$, the replacement $S = I$ is therefore an approximation rather than an exact description.

Check

The overall sign of each eigenvector is arbitrary. For example,

$$\mathbf{c} \quad \text{and} \quad -\mathbf{c}$$

represent the same physical eigenstate. If

$$H\mathbf{c} = ES\mathbf{c},$$

then multiplying the eigenvector by -1 gives

$$H(-\mathbf{c}) = ES(-\mathbf{c}),$$

so the eigenvalue E is unchanged.

For the numerical solution,

$$E_1 = -6.05084746 \text{ eV}, \quad E_2 = -4.31707317 \text{ eV}.$$

The S -orthonormality check gives

$$C^T S C = \begin{pmatrix} 1.00000000 & -1.30893359 \times 10^{-16} \\ 1.17879344 \times 10^{-17} & 1.00000000 \end{pmatrix} \approx I.$$

Thus, although the signs of the eigenvectors may change, the energies remain the same and the eigenvectors remain S -orthonormal.

Problem 2

Question 1

Using $\epsilon = 0$ and $t = 1$, the even-singlet Hamiltonian is

$$H_{\text{even}} = \begin{pmatrix} 0 & -2 \\ -2 & U \end{pmatrix}.$$

The matrix was constructed and diagonalized numerically for

$$U/t = 0, 1, 2, 4, 8, 16.$$

The resulting eigenvalues are

U/t	E_1/t	E_2/t
0	-2.000000	2.000000
1	-1.561553	2.561553
2	-1.236068	3.236068
4	-0.828427	4.828427
8	-0.472136	8.472136
16	-0.246211	16.246211

Question 2

For each value of U/t , the ground-state energy was taken as the lower eigenvalue of H_{even} .

The ground-state eigenvector is written in the basis

$$\{\Phi_C, \Phi_{I+}\}$$

as

$$\begin{pmatrix} c_C \\ c_I \end{pmatrix},$$

so the ionic probability is

$$|c_I|^2.$$

The triplet energy is

$$E_T = 2\epsilon = 0,$$

so the singlet–triplet gap is

$$J = E_T - E_{\text{GS}} = -E_{\text{GS}}.$$

The numerical results are

U/t	E_{GS}/t	$ c_I ^2$	J/t
0	-2.000000	0.500000	2.000000
1	-1.561553	0.378732	1.561553
2	-1.236068	0.276393	1.236068
4	-0.828427	0.146447	0.828427
8	-0.472136	0.052786	0.472136
16	-0.246211	0.014929	0.246211

As U/t increases, the ionic probability decreases, showing that double occupancy becomes less important in the ground state.

Question 3

The numerical ground-state energies were compared with the exact expression

$$E_{\text{GS}} = 2\epsilon + \frac{U}{2} - \sqrt{4t^2 + \frac{U^2}{4}}.$$

Using $\epsilon = 0$ and $t = 1$, the exact energies were evaluated for each assigned value of U/t .

U/t	$E_{\text{numerical}}/t$	E_{exact}/t	Difference
0	-2.000000	-2.000000	0.00×10^0
1	-1.561553	-1.561553	0.00×10^0
2	-1.236068	-1.236068	2.22×10^{-16}
4	-0.828427	-0.828427	1.11×10^{-16}
8	-0.472136	-0.472136	2.22×10^{-16}
16	-0.246211	-0.246211	1.11×10^{-16}

The numerical and exact ground-state energies are essentially the same. The very small differences are due to roundoff.

Question 4

For $U > 0$, the exact singlet-triplet gap

$$J = E_T - E_{\text{GS}}$$

was compared with the large- U superexchange approximation

$$J_{\text{SE}} = \frac{4t^2}{U}.$$

The numerical comparison is

U/t	J_{exact}/t	$4t^2/(Ut)$
1	1.561553	4.000000
2	1.236068	2.000000
4	0.828427	1.000000
8	0.472136	0.500000
16	0.246211	0.250000

The superexchange approximation is bad for small U/t , but becomes increasingly accurate as U/t grows. For example, at $U/t = 16$,

$$J_{\text{exact}}/t = 0.246211, \quad 4t^2/(Ut) = 0.250000,$$

showing that the exact gap is approaching the large- U result

$$J \approx \frac{4t^2}{U}.$$

Question 5

The exact singlet–triplet gap and the ionic probability were evaluated over the range

$$0.25 \leq U/t \leq 20.$$

The exact gap J/t was compared with the large- U superexchange approximation $4t/U$. The ionic probability $|c_I|^2$ was obtained from the ionic component of the ground-state eigenvector.

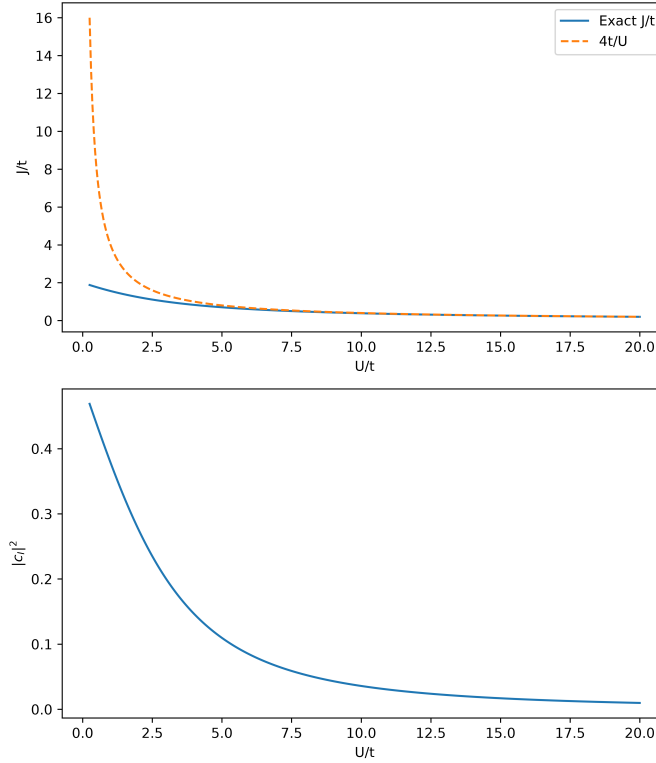


Figure 2: Top: exact singlet–triplet gap J/t compared with the superexchange approximation $4t/U$. Bottom: ionic probability $|c_I|^2$ as a function of U/t .

Question 6

As U/t increases, placing both electrons on the same site becomes more energetically expensive. Therefore, the ionic contribution to the ground state decreases and the double occupancy is suppressed. Thus, the electrons become more localized on separate sites.

For small U/t , the covalent and ionic states mix strongly. This means the large- U superexchange approximation is not very accurate. At large U/t , double

occupancy is strongly suppressed and appears mainly as a virtual intermediate state during hopping. In this regime, the superexchange approximation becomes accurate. The numerical results agree strongly with this.

Problem 3

Question 1

The bond vector is

$$\mathbf{R}_{12} = (3.2, 2.3, 2.6) \text{ \AA}.$$

All calculations were done numerically.

The bond length is

$$d = |\mathbf{R}_{12}| = \sqrt{3.2^2 + 2.3^2 + 2.6^2} = 4.721229 \text{ \AA}.$$

The direction cosines are

$$l = \frac{3.2}{d} = 0.677790,$$

$$m = \frac{2.3}{d} = 0.487161,$$

$$n = \frac{2.6}{d} = 0.550704.$$

The normalization check gives

$$l^2 + m^2 + n^2 = 1.000000,$$

Question 2

Using

$$\frac{\hbar^2}{m_e} = 7.62 \text{ eV \AA}^2$$

and the bond length from Question 1,

$$d = 4.721229 \text{ \AA},$$

we numerically get that

$$V_{ss\sigma} = -1.32 \frac{\hbar^2}{m_e d^2} = -0.451252 \text{ eV},$$

$$V_{sp\sigma} = 1.42 \frac{\hbar^2}{m_e d^2} = 0.485437 \text{ eV},$$

$$V_{pp\sigma} = 2.22 \frac{\hbar^2}{m_e d^2} = 0.758923 \text{ eV},$$

$$V_{pp\pi} = -0.63 \frac{\hbar^2}{m_e d^2} = -0.215370 \text{ eV}.$$

Question 3

Using the direction cosines from Question 1 and the coupling parameters from Question 2, numerically we get that the Hamiltonian in the basis

$$\{s_1, p_{x1}, p_{y1}, p_{z1}\}$$

for site 1 and

$$\{s_2, p_{x2}, p_{y2}, p_{z2}\}$$

for site 2 is

$$H_{12} = \begin{pmatrix} -0.451252 & 0.329024 & 0.236486 & 0.267332 \\ -0.329024 & 0.232219 & 0.321705 & 0.363666 \\ -0.236486 & 0.321705 & 0.015855 & 0.261385 \\ -0.267332 & 0.363666 & 0.261385 & 0.080109 \end{pmatrix} \text{ eV.}$$

The rows correspond to

$$s_1, p_{x1}, p_{y1}, p_{z1},$$

and the columns correspond to

$$s_2, p_{x2}, p_{y2}, p_{z2}.$$

Question 4

Reversing the bond changes

$$\mathbf{R}_{12} \rightarrow -\mathbf{R}_{12},$$

so the direction cosines change as

$$(l, m, n) \rightarrow (-l, -m, -n).$$

The numerical evaluation of the reversed-bond matrix is

$$H_{12}(-\mathbf{R}_{12}) = \begin{pmatrix} -0.451252 & -0.329024 & -0.236486 & -0.267332 \\ 0.329024 & 0.232219 & 0.321705 & 0.363666 \\ 0.236486 & 0.321705 & 0.015855 & 0.261385 \\ 0.267332 & 0.363666 & 0.261385 & 0.080109 \end{pmatrix} \text{ eV.}$$

The numerical evaluation of the transpose of the forward-bond matrix is

$$H_{12}(\mathbf{R}_{12})^T = \begin{pmatrix} -0.451252 & -0.329024 & -0.236486 & -0.267332 \\ 0.329024 & 0.232219 & 0.321705 & 0.363666 \\ 0.236486 & 0.321705 & 0.015855 & 0.261385 \\ 0.267332 & 0.363666 & 0.261385 & 0.080109 \end{pmatrix} \text{ eV.}$$

The maximum numerical difference is

$$\max |H_{12}(-\mathbf{R}_{12}) - H_{12}(\mathbf{R}_{12})^T| = 0.0,$$

so the required relation

$$H_{12}(-\mathbf{R}_{12}) = H_{12}(\mathbf{R}_{12})^T$$

is correct to numerical precision.

Question 5

The matrix elements that change sign under bond reversal are the s - p couplings:

$$H_{s_1, p_{x2}}, \quad H_{s_1, p_{y2}}, \quad H_{s_1, p_{z2}},$$

and

$$H_{p_{x1}, s_2}, \quad H_{p_{y1}, s_2}, \quad H_{p_{z1}, s_2}.$$

Numerically,

$$0.329024 \rightarrow -0.329024,$$

$$0.236486 \rightarrow -0.236486,$$

$$0.267332 \rightarrow -0.267332,$$

with the corresponding reverse p - s elements changing sign in the opposite direction.

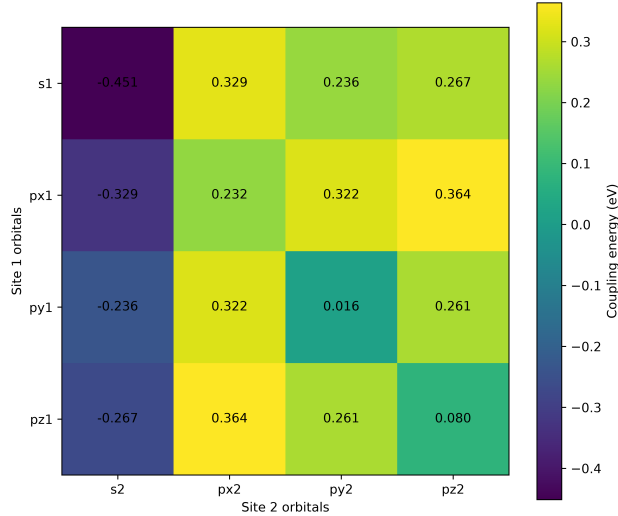


Figure 3: Heat map of the coupling matrix $H_{12}(\mathbf{R}_{12})$. Each square represents the coupling between one orbital on site 1 and one orbital on site 2. The color indicates the value and sign of the corresponding Hamiltonian matrix element.

The heat map shows how the bond geometry affects the strength and sign of the different orbital-to-orbital couplings.

Question 6

The s - p matrix elements are odd under bond reversal because when reversing, the bond changes

$$(l, m, n) \rightarrow (-l, -m, -n),$$

so terms such as

$$lV_{sp\sigma}, \quad mV_{sp\sigma}, \quad nV_{sp\sigma}$$

change sign.

The p - p matrix elements are even under bond reversal because its direction depends on products such as

$$l^2, \quad m^2, \quad n^2, \quad lm, \quad ln, \quad mn.$$

These remain unchanged when all direction cosines reverse sign. Therefore, the s - p elements change sign while the p - p elements do not.

Problem 4

Question 1

Using the ordered basis

$$\{s_1, p_{x1}, p_{y1}, p_{z1}, s_2, p_{x2}, p_{y2}, p_{z2}\},$$

for each oxygen atom we get

$$D = \begin{pmatrix} \epsilon_s & 0 & 0 & 0 \\ 0 & \epsilon_p & 0 & 0 \\ 0 & 0 & \epsilon_p & 0 \\ 0 & 0 & 0 & \epsilon_p \end{pmatrix},$$

with

$$\epsilon_s = -34.02 \text{ eV}, \quad \epsilon_p = -16.77 \text{ eV}.$$

Since the molecular axis is along the z direction,

$$(l, m, n) = (0, 0, 1),$$

so the Slater–Koster Hamiltonian is

$$H_{12} = \begin{pmatrix} V_{ss\sigma} & 0 & 0 & V_{sp\sigma} \\ 0 & V_{pp\pi} & 0 & 0 \\ 0 & 0 & V_{pp\pi} & 0 \\ -V_{sp\sigma} & 0 & 0 & V_{pp\sigma} \end{pmatrix}.$$

The complete molecular Hamiltonian is therefore

$$H = \begin{pmatrix} D & H_{12} \\ H_{12}^T & D \end{pmatrix}.$$

The Harrison matrix elements were evaluated numerically as functions of bond distance using

$$V_{ss\sigma} = -1.32 \frac{\hbar^2}{m_e d^2},$$

$$V_{sp\sigma} = 1.42 \frac{\hbar^2}{m_e d^2},$$

$$V_{pp\sigma} = 2.22 \frac{\hbar^2}{m_e d^2},$$

and

$$V_{pp\pi} = -0.63 \frac{\hbar^2}{m_e d^2},$$

with

$$\frac{\hbar^2}{m_e} = 7.62 \text{ eV } \text{\AA}^2.$$

The Hamiltonian was diagonalized for several O–O separations. For every distance,

$$\max |H - H^T| = 0,$$

confirming that the Hamiltonian is Hermitian.

The calculated molecular-orbital energies and classifications are:

d (Å)	State	E (eV)	Type
1.000	1	-50.885711	σ
	2	-28.105747	σ^*
	3	-26.879089	σ
	4	-21.570600	π
	5	-21.570600	π
	6	-11.969400	π^*
	7	-11.969400	π^*
	8	4.290547	σ^*
1.208	1	-44.351499	σ
	2	-29.397332	σ^*
	3	-24.923710	σ
	4	-20.059741	π
	5	-20.059741	π
	6	-13.480259	π^*
	7	-13.480259	π^*
	8	-2.907459	σ^*
1.500	1	-39.965607	σ
	2	-30.631331	σ^*
	3	-22.813193	σ
	4	-18.903600	π
	5	-18.903600	π
	6	-14.636400	π^*
	7	-14.636400	π^*
	8	-8.169869	σ^*
2.000	1	-36.992147	σ
	2	-31.883709	σ^*
	3	-20.541553	σ
	4	-17.970150	π
	5	-17.970150	π
	6	-15.569850	π^*
	7	-15.569850	π^*
	8	-12.162591	σ^*

The π states occur in degenerate pairs because the p_x and p_y orbitals are equivalent for a bond oriented along z . The σ states contain contributions from the s and p_z orbitals, while the π states contain contributions from the p_x and p_y orbitals.

Question 2

Each oxygen atom contributes six valence electrons, so the O_2 molecule contains

$$N_e = 12$$

valence electrons.

The molecular orbitals are filled from lowest to highest energy. By the Pauli exclusion principle, each orbital can contain at most two electrons with opposite spins. For the two degenerate π^* orbitals, Hund's rule requires the electrons to occupy the two orbitals singly before pairing.

The occupations are

State	Type	Occupation
1	σ	2
2	σ^*	2
3	σ	2
4	π	2
5	π	2
6	π^*	1
7	π^*	1
8	σ^*	0

Thus there are eight electrons in bonding orbitals and four electrons in antibonding orbitals:

$$N_b = 8, \quad N_a = 4.$$

The bond order is therefore

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{8 - 4}{2} = \boxed{2}.$$

The model therefore predicts a double bond for O_2 . The two highest occupied π^* orbitals are each singly occupied, leaving two unpaired electrons.

Question 3

The separated-atom energy is

$$E_{\text{sep}} = 2(2\epsilon_s + 4\epsilon_p) = -270.24 \text{ eV}.$$

Using the occupations from Question 2,

$$E_{\text{band}}(d) = \sum_n f_n \epsilon_n(d).$$

At

$$d_0 = 1.208 \text{ \AA},$$

numerically we get that the band energy is

$$E_{\text{band}}(d_0) = -304.544564 \text{ eV}.$$

The derivative of the band energy at d_0 was evaluated numerically using the centered finite difference

$$\left. \frac{dE_{\text{band}}}{dd} \right|_{d_0} \approx \frac{E_{\text{band}}(d_0 + h) - E_{\text{band}}(d_0 - h)}{2h}.$$

This gives

$$\left. \frac{dE_{\text{band}}}{dd} \right|_{d_0} = 61.690623 \text{ eV/\AA}.$$

The total energy is

$$E_{\text{tot}}(d) = E_{\text{band}}(d) + \frac{C}{d^4}.$$

The constant C is determined by requiring

$$\left. \frac{dE_{\text{tot}}}{dd} \right|_{d_0} = 0.$$

Therefore,

$$0 = \left. \frac{dE_{\text{band}}}{dd} \right|_{d_0} - \frac{4C}{d_0^5},$$

so

$$C = \frac{d_0^5}{4} \left. \frac{dE_{\text{band}}}{dd} \right|_{d_0}.$$

Numerically,

$$\boxed{C = 39.672890 \text{ eV \AA}^4}.$$

At $d_0 = 1.208 \text{ \AA}$, the repulsive contribution is

$$V_{\text{rep}}(d_0) = \frac{C}{d_0^4} = 18.630568 \text{ eV}.$$

The total energy is therefore

$$E_{\text{tot}}(d_0) = E_{\text{band}}(d_0) + V_{\text{rep}}(d_0) = -285.913996 \text{ eV}.$$

Finally,

$$\Delta E(d_0) = E_{\text{tot}}(d_0) - E_{\text{sep}},$$

which gives

$$\boxed{\Delta E(d_0) = -15.673996 \text{ eV}}.$$

Since $\Delta E(d_0) < 0$, the bonded O_2 configuration is lower in energy than the two separated oxygen atoms in this model.

Question 4

A fine grid of O–O separations was used to search for the minimum of the total energy,

$$E_{\text{tot}}(d) = E_{\text{band}}(d) + \frac{C}{d^4}.$$

The minimum occurs at

$$d_{\text{eq}} = 1.208 \text{ \AA}.$$

At this distance,

$$E_{\text{tot}}(d_{\text{eq}}) = -285.913996 \text{ eV}.$$

The cohesive energy is

$$E_{\text{coh}} = E_{\text{sep}} - E_{\text{tot}}(d_{\text{eq}}).$$

Using

$$E_{\text{sep}} = -270.24 \text{ eV},$$

gives

$$E_{\text{coh}} = -270.24 - (-285.913996),$$

so

$$E_{\text{coh}} = 15.673996 \text{ eV}.$$

The numerical minimum at 1.208 Å confirms the equilibrium distance used to calibrate the repulsive interaction in Question 3.

Question 5

The three energy contributions were plotted as functions of the O–O separation d :

$$E_{\text{band}}(d) - E_{\text{sep}},$$

$$V_{\text{rep}}(d) = \frac{C}{d^4},$$

and

$$\Delta E(d) = E_{\text{tot}}(d) - E_{\text{sep}}.$$

Since

$$E_{\text{tot}}(d) = E_{\text{band}}(d) + V_{\text{rep}}(d),$$

the total energy difference can also be written as

$$\Delta E(d) = [E_{\text{band}}(d) - E_{\text{sep}}] + V_{\text{rep}}(d).$$

The electronic band-energy contribution lowers the energy and favors bonding, while the repulsive contribution increases rapidly at short distances. Their sum produces the total binding-energy curve with a minimum near the equilibrium separation.

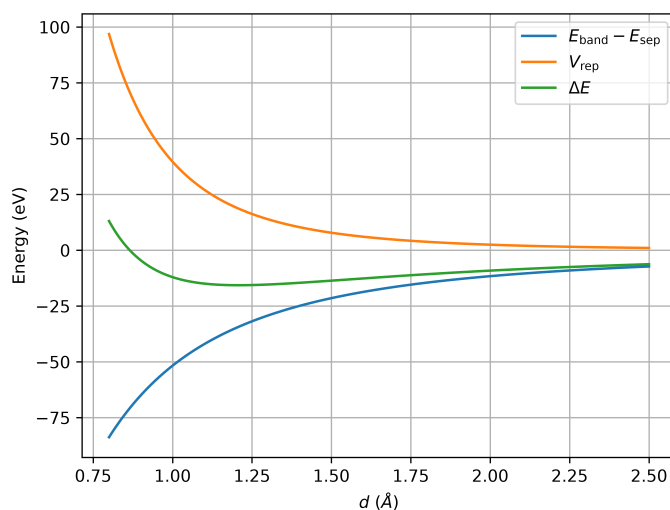


Figure 4: Electronic band-energy contribution, repulsive contribution, and total energy difference as functions of the O–O separation.

Question 6

From the molecular-orbital filling in Question 2, the two degenerate π^* orbitals are each occupied by one electron:



Therefore, the molecule contains

2 unpaired electrons

Because these unpaired electrons produce a net magnetic moment, the molecule is

paramagnetic

Question 7

The tight-binding model reproduces several important properties of O_2 .

The calculated bond order is

$$\boxed{2},$$

which agrees with the usual description of O_2 as having an O=O double bond.

The equilibrium bond length is

$$\boxed{d_{\text{eq}} = 1.208 \text{ \AA}}.$$

This is consistent with the known O–O bond length.

The model gives a cohesive energy of

$$\boxed{E_{\text{coh}} = 15.673996 \text{ eV}}.$$

The positive cohesive energy correctly indicates that the bonded molecule is more stable than two separated oxygen atoms, even though it overestimate this energy compared to what is known.

Finally, the model gives two unpaired electrons in the degenerate π^* orbitals. Therefore it predicts that O_2 is paramagnetic, which agrees the known magnetic behavior of oxygen.

Two important limitations are:

- The one-electron tight-binding model does not fully include electron-electron interactions and electron correlation, which affects the calculated molecular energies.
- The repulsive interaction,

$$V_{\text{rep}} = \frac{C}{d^4},$$

uses C which was fitted using the known bond length. Therefore the calculated bond length is not an independent prediction of the model.

Python Documentation

For each problem, a documented Python source files used for the numerical calculations and figures will be submitted separately with the assignment.